

Module :- Windows Server Security and Maintenance

31. Explain the process of installing and configuring Hyper-V virtualization in Windows Server 2016.

Ans:- **Hyper-V** is Microsoft's virtualization platform used to create and manage virtual machines (VMs).

Installation Steps:

1. Open **Server Manager**
2. Click **Manage → Add Roles and Features**
3. Select **Role-based or feature-based installation**
4. Choose the target server
5. Select **Hyper-V** from server roles
6. Click **Add Features**
7. Select virtual switch options (if needed)
8. Choose VM storage location
9. Click **Install**
10. Restart the server after installation

Configuration Steps:

1. Open **Hyper-V Manager**
2. Create **Virtual Switch** for network access
3. Click **New → Virtual Machine**
4. Set:
 - VM Name
 - Generation
 - RAM
 - Network Adapter
 - Virtual Hard Disk
 - ISO file for OS installation
5. Start and manage the VM

32. How do you monitor server performance and manage event logs in Windows Server?

Ans:- **Performance Monitoring:**

Use:

- **Task Manager**
- **Performance Monitor (perfmon)**
- **Resource Monitor**
- **Server Manager**

Monitor:

- CPU usage
- RAM usage
- Disk activity
- Network traffic

Event Log Management:

Use **Event Viewer**

Steps:

1. Open **Server Manager** → **Tools** → **Event Viewer**
2. Check logs:
 - Application
 - Security
 - System
 - Setup
3. Filter logs for warnings/errors
4. Review Event ID and details
5. Take corrective actions

33. Describe the different types of storage options available in Windows Server.

Ans:-

Types of Storage:

1. DAS (Direct Attached Storage)

Storage directly connected to server

2. NAS (Network Attached Storage)

Storage accessed through network

3. SAN (Storage Area Network)

High-speed block-level storage network

4. Local Disks

HDD, SSD, NVMe drives

5. Storage Spaces

Combines multiple disks into storage pools

6. iSCSI Storage

Access remote disks over IP network

7. Cloud Storage

Using services like Microsoft Azure

34. What is the role of File Server in Windows Server, and how do you configure it?

Ans:- **Role of File Server:**

Used to:

- Store files centrally
- Share files over network
- Control access permissions
- Enable data security

Configuration Steps:

1. Open **Server Manager**
2. Add Role → **File and Storage Services**
3. Create a folder (example: D:\Shared)
4. Right-click → Properties → Sharing

5. Enable **Advanced Sharing**
6. Set share name
7. Configure:
 - Share permissions
 - NTFS permissions
8. Access using:
\\ServerName\Shared

35. Explain the process of implementing and managing Distributed File System (DFS) in Windows Server 2016.

Ans:- **DFS (Distributed File System)** allows shared folders from different servers to appear as one folder.

Steps:

1. Install **DFS Namespaces** and **DFS Replication**
2. Open **DFS Management**
3. Create **New Namespace**
4. Select server and namespace name
5. Add shared folders (folder targets)
6. Configure replication between servers
7. Set replication schedule and bandwidth

Benefits:

- Centralized access
- Fault tolerance
- Replication between servers

36. Discuss the built-in backup and recovery options available in Windows Server 2016 or 2019.

Ans:-

Backup Options:

- Windows Server Backup
- System State Backup
- Bare Metal Recovery
- Full Server Backup
- Volume Backup
- File and Folder Backup

Recovery Options:

- File Recovery
- System Restore
- Bare Metal Recovery
- Active Directory Restore
- Windows Recovery Environment (WinRE)

37. How do you configure Windows Server Backup to back up critical data?

Ans:-

Steps:

1. Open **Server Manager**
2. Add Feature → **Windows Server Backup**
3. Open **Windows Server Backup**
4. Click **Local Backup → Backup Schedule**
5. Choose:
 - Full server
 - Custom backup
6. Select backup time
7. Choose destination:
 - Local disk
 - External drive
 - Network shared folder

8. Finish configuration

38. Explain the steps for restoring files and folders using Windows Server Backup.

Ans:-

Steps:

1. Open **Windows Server Backup**
2. Click **Local Backup → Recover**
3. Select backup location
4. Choose backup date/time
5. Select recovery type:
 - Files and folders
6. Browse and select files
7. Choose restore location:
 - Original location
 - Alternate location
8. Click **Recover**

39. What are some common troubleshooting techniques for Windows Server startup issues?

Ans:-

Common Techniques:

1. Safe Mode

Boot with minimal drivers

2. Last Known Good Configuration

Restore previous working settings

3. Startup Repair

Use Windows Recovery tools

4. Check Boot Configuration

Use:

bootrec /fixmbr

bootrec /fixboot

bootrec /rebuildbcd

5. Verify Disk Health

Use:

chkdsk /f /r

6. Check Event Logs

Use Event Viewer for startup errors

7. Driver Troubleshooting

Rollback problematic drivers

40. How do you troubleshoot network connectivity problems in Windows Server?

Ans:-

Steps:

1. Check Physical Connection

Cable, switch, NIC status

2. Verify IP Configuration

Use:

ipconfig /all

3. Test Connectivity

Use:

ping

tracert

pathping

4. DNS Troubleshooting

Use:

nslookup

ipconfig /flushdns

5. Check Firewall Rules

Windows Firewall may block traffic

6. Restart Network Adapter

Disable/enable NIC

7. Verify DHCP/DNS Services

41. Discuss common Active Directory-related issues and their troubleshooting steps.

Ans:-

Common Issues:

- User login failure
- Replication problems
- DNS issues
- Group Policy not applying
- Domain controller unavailable

Troubleshooting Steps:

1. Check DNS

AD depends heavily on DNS

2. Verify Replication

Use:

repadmin /replsummary

3. Check Domain Controller Health

Use:

dcdiag

4. Verify Time Synchronization

Kerberos requires correct time

5. Check SYSVOL and NETLOGON Shares

6. Force Group Policy Update

Use:

gpupdate /force

42. Explain how to troubleshoot performance problems on Windows Server 2016 or 2019

Ans:-

Common Performance Issues:

- High CPU usage
- Low memory
- Disk bottlenecks
- Slow network
- Application hangs

Troubleshooting Steps:

1. Use Performance Monitor

Check counters like:

- Processor
- Memory
- Disk Queue Length
- Network usage

2. Check Task Manager

Identify heavy processes

3. Review Event Viewer

Find warnings/errors

4. Analyze Disk Usage

Check disk fragmentation and space

5. Update Drivers and Windows Updates

6. Disable Unnecessary Services

7. Check Antivirus Impact

8. Monitor Resource Bottlenecks

This helps improve system stability and performance.

